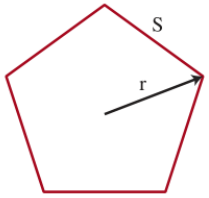
Started on	Wednesday, 11 March 2026, 10:03 PM
State	Finished
Completed on	Wednesday, 11 March 2026, 10:05 PM
Time taken	2 mins 9 secs
Grade	**0.00** out of 100.00

Question 1

Not answered

Mark 0.00 out of 15.00

Write a JAVA program that prompts the user to enter the length from the center of a pentagon to a vertex and computes the area of the pentagon, as shown in the following figure.



The formula for computing the area of a pentagon is $Area = \frac{5 \times s^2}{4 \times \tan\left(\frac{\pi}{5}\right)}$ where ***s*** is the length of a side. The side can be computed using the formula $s = 2r \sin\frac{\pi}{5}$, where ***r*** is the length from the center of a pentagon to a vertex.

Hint: Round up two digits after the decimal point.

For example:

Input	Result
5.5	Enter the length from the center to a vertex: 5.5 The area of the pentagon is 71.92
12.1	Enter the length from the center to a vertex: 12.1 The area of the pentagon is 348.11

Answer: (penalty regime: 0 %)

```
1
```

▼ **Show/hide question author's solution (Java)**

```java
import java.util.Scanner;

public class Exercise04_01 {
  public static void main(String[] args) {
    Scanner input = new Scanner(System.in);

    System.out.print("Enter the length from the center to a vertex: ");
    double r = input.nextDouble();
    System.out.println(r);

```

```
    double side = 2 * r * Math.sin(Math.PI / 5);
    double area = 5 * side * side / (4 * Math.tan(Math.PI / 5));

    System.out.println("The area of the pentagon is " + Math.round(area * 100) / 100.0);
  }
}
```

Question 2

Not answered

Mark 0.00 out of 15.00

Write a program that prompts the user to enter a letter and check whether the letter is a vowel or consonant.

For a non-letter input, display "**Invalid input**".

Hint: You can use ***toUpperCase()*** method to check both capital and small letters.

For example:

Input	Result
a	Enter a letter: a a is a vowel
Z	Enter a letter: Z Z is a consonant
#	Enter a letter: # # is an invalid input

Answer: (penalty regime: 0 %)

```
1
```

Show/hide question author's solution (Java)

```java
import java.util.Scanner;

public class Exercise04_13 {
  public static void main(String[] args) {
    Scanner input = new Scanner(System.in);
    System.out.print("Enter a letter: ");
    char letter = input.nextLine().charAt(0);
    System.out.println(letter);

    if (Character.toUpperCase(letter) == 'A' || Character.toUpperCase(letter) == 'E' || Character.toUpperCase(lette
        || Character.toUpperCase(letter) == 'O' || Character.toUpperCase(letter) == 'U')
      System.out.println(letter + " is a vowel");
    else if (Character.isLetter(letter))
      System.out.println(letter + " is a consonant");
    else
      System.out.println(letter + " is an invalid input");
  }
}
```

Question 3

Not answered

Mark 0.00 out of 15.00

Write a program that prompts the user to enter three cities and displays them in ascending order.

For example:

Input	Result
Bursa Izmit Izmir	Enter the first city: Bursa Enter the second city: Izmit Enter the third city: Izmir The three cities in alphabetical order are Bursa Izmir Izmit
California NewYork NewJersey	Enter the first city: California Enter the second city: NewYork Enter the third city: NewJersey The three cities in alphabetical order are California NewJersey NewYork

Answer: (penalty regime: 0 %)

```
1
```

Show/hide question author's solution (Java)

```java
import java.util.Scanner;

public class Exercise04_24 {
  public static void main(String[] args) {
    Scanner input = new Scanner(System.in);
    System.out.print("Enter the first city: ");
    String s1 = input.nextLine();
    System.out.println(s1);
    System.out.print("Enter the second city: ");
    String s2 = input.nextLine();
    System.out.println(s2);
    System.out.print("Enter the third city: ");
    String s3 = input.nextLine();
    System.out.println(s3);

    if (s1.compareTo(s2) > 0) {
      // Swap s1 with s2
      String temp = s1;
      s1 = s2;
      s2 = temp;
    }

    if (s2.compareTo(s3) > 0) {
      // Swap s2 with s3
      String temp = s2;
      s2 = s3;
      s3 = temp;
```

```
28      }
29
30      if (s1.compareTo(s2) > 0) {
31        // Swap s1 with s2
32        String temp = s1;
33        s1 = s2;
34        s2 = temp;
35      }
36
37      System.out.print("The three cities in alphabetical order are "
38        + s1 + " " + s2 + " " + s3);
39    }
40  }
```

Question 4

Not answered

Mark 0.00 out of 15.00

Write a program that reads an unspecified number of integers, determines how many positive and negative values have been read, and computes the total and average of the input values (not counting zeros).

Your program ends with the input **0**.

Display the average as a floating-point number.

For example:

Input	Result
5 15 -56 223 19 0	Enter an integer, the input ends if it is 0: 5 15 -56 223 19 0 The number of positives is 4 The number of negatives is 1 The total is 206 The average is 41.2
78 1584 -65 0	Enter an integer, the input ends if it is 0: 78 1584 -65 0 The number of positives is 2 The number of negatives is 1 The total is 1597 The average is 532.3333333333334

Answer: (penalty regime: 0 %)

```
1
```

Show/hide question author's solution (Java)

```java
import java.util.Scanner;

public class Exercise05_01 {
  public static void main(String[] args) {
    int countPositive = 0, countNegative = 0;
    int count = 0, total = 0, number;

    Scanner input = new Scanner(System.in);
    System.out.print("Enter an integer, the input ends if it is 0: ");
    number = input.nextInt();
    System.out.print(number);

    while (number != 0) {
      if (number > 0)
        countPositive++;
      else if (number < 0)
        countNegative++;

      total += number;
      count++;
```

```
      // Read the next number
      number = input.nextInt();
      System.out.print(" " + number);
    }

    if (count == 0)
      System.out.println("No numbers are entered except 0");
    else {
      System.out.println("\nThe number of positives is " + countPositive);
      System.out.println("The number of negatives is " + countNegative);
      System.out.println("The total is " + total);
      System.out.println("The average is " + total * 1.0 / count);
    }
  }
}
```

Question 5

Not answered

Mark 0.00 out of 20.00

Write a program that prompts the user to enter two positive integers and displays the gcd (great common divisor).

Hint: A solution to find the greatest common divisor of two integers **n1** and **n2** is as follows:

First find **d** to be the minimum of **n1** and **n2**, then check whether **d, d–1, d–2, ..., 2, or 1** is a divisor for both **n1** and **n2** in this order.

The **first** such common divisor is the greatest common divisor for **n1** and **n2**.

For example:

Input	Result
65 175	Enter the first number: 65 Enter the second number: 175 GCD of 65 and 175 is 5
844 16	Enter the first number: 844 Enter the second number: 16 GCD of 844 and 16 is 4

Answer: (penalty regime: 0 %)

```
1
```

Show/hide question author's solution (Java)

```java
public class Exercise05_14 {
  public static void main(String[] args) {
    java.util.Scanner input = new java.util.Scanner(System.in);

    // Enter n1
    System.out.print("Enter the first number: ");
    int n1 = input.nextInt();
    System.out.println(n1);
    // Enter n2
    System.out.print("Enter the second number: ");
    int n2 = input.nextInt();
    System.out.println(n2);
    int d = (n1 < n2) ? n1 : n2;
    for (; d >= 1; d--) {
      if ((n1 % d == 0) && (n2 % d == 0)) {
        break;
      }
    }

    System.out.println("GCD of " + n1 + " and " + n2 + " is " + d);
  }
}

```

Question 6

Not answered

Mark 0.00 out of 20.00

Write a program that prompts the user to enter a string and displays the string in reverse order.

For example:

Input	Result
AbCDDEdg	Enter a string: AbCDDEdg The reversed string is gdEDDCbA
APtHueR	Enter a string: APtHueR The reversed string is ReuHtPA

Answer: (penalty regime: 0 %)

```
1
```

Show/hide question author's solution (Java)

```java
import java.util.Scanner;

public class Exercise05_46 {
  public static void main(String[] args) {
    Scanner input = new Scanner(System.in);
    System.out.print("Enter a string: ");
    String s = input.nextLine();
    System.out.println(s);

    String result = "";
    for (int i = 0; i < s.length(); i++) {
      result = s.charAt(i) + result;
    }

    System.out.println("The reversed string is " + result);
  }
}
```